IDX Exchange

Data Analyst Internship

MLS Analytics & Tableau Dashboard Program

Intern Handbook

12-Week Program Guide

Confidential – For Intern Use Only

Welcome to IDX Exchange

Congratulations on joining the IDX Exchange Data Analyst Internship Program. Over the next 12 weeks, you will gain hands-on experience analyzing real MLS transaction data, building market analytics dashboards in Tableau, and communicating meaningful housing market insights.

This handbook is your complete reference guide for the program. Keep it handy throughout the internship – it outlines every week's objectives, the skills you will develop, what is expected of you at each stage, and the deliverable due at the end of each phase.

Program Mission

Teach students how real estate analytics datasets are produced, cleaned, and transformed into actionable housing market intelligence using industry-standard tools.

Program Overview

The internship is structured as a progressive pipeline – each phase of the program builds directly on the last:

Phase	Description
Data Cleaning	Prepare raw data for reliable analysis
Market Analytics	Engineer key housing market metrics
Competitive Intelligence	Identify top agents and brokerages
Dashboard Development	Build interactive Tableau dashboards
Market Insights	Communicate findings through reports and presentations

Program Details

Detail	Information
Duration	12 Weeks (+ Week 0 Orientation)
Primary Tools	Python (Pandas), Tableau Desktop
Data Source	CoreLogic Trestle API via IDX Exchange pipeline
Dataset Types	Monthly MLS Listing & Sold transaction CSV files
Final Deliverable	Tableau Dashboards + 1-Page Market Intelligence Report + Presentation

Required Software Setup

Before your first week begins, please download and install the following tools on your computer.

1. Python IDE (choose one)

You will need a Python coding environment throughout this internship. Download and install one of the following:

- Spyder – Scientific Python development environment, great for data analysis (included with Anaconda). Download at: anaconda.com
- VS Code (Visual Studio Code) – Lightweight and widely used editor with excellent Python support. Download at: code.visualstudio.com
- IDE of your choice – If you already use a preferred Python environment (e.g., PyCharm, Jupyter Lab), you are welcome to continue using it.

2. Tableau Desktop Public Edition (Free)

Tableau Desktop Public Edition is the free version of Tableau and is the primary visualization tool used in Weeks 8–12. Download it at tableau.com/products/public/download and create a free account.

Week-by-Week Curriculum

Week 0 – MLS Data Pipeline Orientation

Before analytics begins, you will be introduced to how the MLS datasets used throughout this program are generated. Understanding the data's origins is essential context for everything that follows.

Objective

Understand how monthly MLS datasets are produced from the CoreLogic Trestle API and exported as CSV files for analysis.

Topics Covered

- API authentication and token management
- Pagination handling for large dataset extraction
- Filtering listing and sold records by date ranges
- Exporting monthly datasets to structured CSV format
- Review Trestle Property Metadata (field definitions, data types, and available property attributes)

Example Pipeline Outputs

```
CRMLSListingYYYYMM.csv
CRMLSSoldYYYYMM.csv
```

Note: Interns may run the extraction script once to observe how datasets are produced. The datasets used throughout Weeks 1–12 are pre-generated and provided at program start.

> **Deliverable – Week 0**
> Run the extraction scripts crmls_sold.py and crmls_listed.py to pull data for any additional months needed. Read the Trestle Property Metadata document to familiarize yourself with field definitions, data types, and available property attributes.

Week 1 – Monthly Dataset Aggregation

Individual monthly files are combined into unified datasets that span multiple months, enabling trend analysis over time.

Objective

Load and concatenate all monthly MLS files from January 2024 through the most recently completed calendar month into analysis-ready combined datasets.

Example Workflow

```
sold1 = pd.read_csv('CRMLSSold202401.csv')
sold2 = pd.read_csv('CRMLSSold202402.csv')
sold = pd.concat([sold1, sold2])
```

Outputs

- Combined sold transactions dataset
- Combined listing data dataset

Skills Learned

- Multi-file dataset management
- Data aggregation with Pandas
- Preparing time-series datasets for analysis

> **Deliverable – Week 1**
> Submit a .py script that concatenates all monthly files from January 2024 through the most recently completed calendar month into two combined datasets (listings and sold), filters both to **PropertyType == 'Residential'** only, and saves them as new CSVs. Include comments confirming row counts before and after concatenation and before and after the Residential filter.

Weeks 2–3 – Dataset Structuring and Validation

Before analytics begins, the dataset must be inspected and filtered to ensure only relevant residential property records are used. This week also covers foundational EDA steps that inform every phase that follows.

Dataset Understanding

- Identify number of rows and columns
- Review column data types
- Identify high-missing columns
- Separate market analysis fields from metadata fields

Missing Value Analysis

- Calculate missing counts and percentages per column
- Flag columns with >90% missing values
- Decide which columns to drop vs. retain (keep core fields even if partially missing)

Numeric Distribution Review

Analyze the distribution of key numeric fields: ClosePrice, ListPrice, OriginalListPrice, LivingArea, LotSizeAcres, BedroomsTotal, BathroomsTotalInteger, DaysOnMarket, and YearBuilt. For each field, generate histograms, boxplots, and percentile summaries, and identify extreme outliers for later handling.

Suggested Intern Questions

Use EDA to answer these questions about your dataset before moving to cleaning:

- What is the Residential vs. other property type share?
- What are the median and average close prices?

- What does the Days on Market distribution look like?
- What percentage of homes sold above vs. below list price?
- Are there any apparent date consistency issues (e.g., close date before listing date)?
- Which counties have the highest median prices?

Tasks

```
# Inspect structure
sold.columns
sold.head()
# Check property categories
sold['PropertyType'].unique()
# Filter residential
sold = sold[sold.PropertyType == 'Residential']
# Validate completeness
sold.isnull().sum()
```

Skills Learned

- Dataset validation and quality checks
- Exploratory data analysis (EDA)
- MLS dataset structure and property type filtering

Deliverable – Weeks 2–3

Submit a .py script documenting unique property types found, the filtering logic applied, and a null-count summary table. Include a missing value report flagging any columns above 90% null. Produce a numeric distribution summary (min, max, mean, median, percentiles) for ClosePrice, LivingArea, and DaysOnMarket. Save the filtered dataset as a new CSV.

Mortgage Rate Enrichment

Enrich both the combined sold and listings datasets by merging in the national 30-year fixed mortgage rate from the St. Louis Federal Reserve (FRED). This introduces interns to fetching live economic data from a public API and performing a time-series join on a monthly key.

Objective

Fetch the FRED MORTGAGE30US series, resample it from weekly to monthly frequency, and merge it onto both combined datasets using a year-month key derived from transaction dates.

Background

The FRED MORTGAGE30US series is published weekly (every Thursday) by Freddie Mac via the St. Louis Federal Reserve. Because MLS transactions are analyzed by calendar month, interns must resample the weekly data to a monthly average before joining. The data can be fetched directly from FRED as a CSV—no API key required.

Tasks

```
# Step 1 – Fetch the mortgage rate data from FRED
import pandas as pd

url = "https://fred.stlouisfed.org/graph/fredgraph.csv?id=MORTGAGE30US"
mortgage = pd.read_csv(url, parse_dates=['observation_date'])
mortgage.columns = ['date', 'rate_30yr_fixed']

# Step 2 – Resample weekly rates to monthly averages
mortgage['year_month'] = mortgage['date'].dt.to_period('M')
mortgage_monthly = (
    mortgage.groupby('year_month')['rate_30yr_fixed']
    .mean()
    .reset_index()
)

# Step 3 – Create a matching year_month key on the MLS datasets

# Sold dataset — key off CloseDate
sold['year_month'] = pd.to_datetime(sold['CloseDate']).dt.to_period('M')

# Listings dataset — key off ListingContractDate
listings['year_month'] = pd.to_datetime(
    listings['ListingContractDate']
).dt.to_period('M')

# Step 4 – Merge
sold_with_rates = sold.merge(mortgage_monthly, on='year_month', how='left')
listings_with_rates = listings.merge(mortgage_monthly, on='year_month', how='left')

# Step 5 – Validate the merge

# Check for any unmatched rows (rate should not be null)
print(sold_with_rates['rate_30yr_fixed'].isnull().sum())
print(listings_with_rates['rate_30yr_fixed'].isnull().sum())

# Preview

print(
    sold_with_rates[
        ['CloseDate', 'year_month', 'ClosePrice', 'rate_30yr_fixed']
    ].head()
)
```

Skills Learned

- Fetching live data from a public API (FRED)
- Resampling time-series data from weekly to monthly frequency
- Creating join keys from datetime fields
- Left merging external economic data onto a transaction dataset
- Validating merge completeness with null checks

> **Deliverable – Weeks 2–3 (continued)**
> Submit a .py script that: (1) fetches the MORTGAGE30US series directly from FRED, (2) resamples it to monthly averages, (3) merges it onto both the combined sold and listings datasets using a year_month key, and (4) includes a validation check confirming no null rate values exist after the merge. Save both enriched datasets as new CSVs.

Weeks 4–5 – Data Cleaning and Preparation

Raw MLS data contains formatting inconsistencies, missing values, and fields that need transformation before analysis. This phase prepares the dataset for reliable analytics.

Tasks

- Convert date fields to datetime format (CloseDate, PurchaseContractDate, ListingContractDate, ContractStatusChangeDate)
- Remove unnecessary or redundant columns
- Handle missing values appropriately
- Ensure numeric fields are properly typed
- Remove or flag invalid numeric values: ClosePrice <= 0, LivingArea <= 0, DaysOnMarket < 0, negative Bedrooms or Bathrooms

Date Consistency Checks

Validate the logical order of date fields: ListingContractDate should precede PurchaseContractDate, which should precede CloseDate. Create boolean flag columns to mark records that violate these rules:

- `listing_after_close_flag`
- `purchase_after_close_flag`
- `negative_timeline_flag`

Geographic Data Checks

- Flag records with missing coordinates (Latitude or Longitude is null)
- Flag Latitude = 0 or Longitude = 0 (sentinel null values)
- Flag Longitude > 0 errors (California coordinates should be negative)
- Flag out-of-state or implausible coordinates

Example

```
sold['CloseDate'] = pd.to_datetime(sold['CloseDate'])
```

Skills Learned

- Data cleaning and transformation
- Preparing analysis-ready datasets
- Working with datetime fields in Pandas

> **Deliverable – Weeks 4–5**
> Submit a cleaned, analysis-ready dataset as a CSV, plus a .py script documenting every transformation made and why. Include before/after row counts, data type confirmations, date consistency flag counts, and a geographic data quality summary noting any invalid coordinate records.

Week 6 – Feature Engineering and Market Metrics

With clean data in hand, you will engineer the key market indicators that power the Tableau dashboards.

Key Metrics to Create

Metric	Formula	Purpose
Price Ratio	ClosePrice / OriginalListPrice	Measures negotiation strength
Price Per Sq Ft	ClosePrice / LivingArea	Normalizes price across sizes
Days on Market	DaysOnMarket (raw field)	Time-to-sell indicator
Year / Month / YrMo	Derived from CloseDate	Enables time-series analysis
Close to Original List Ratio	ClosePrice / OriginalListPrice	Captures full price reduction history
Listing to Contract Days	PurchaseContractDate - ListingContractDate	Measures time from listing to accepted offer
Contract to Close Days	CloseDate - PurchaseContractDate	Escrow and closing period duration

Segment Analysis

After creating your engineered metrics, group the analysis by key dimensions to uncover market patterns. Generate summary statistics for each segment:

- PropertyType and PropertySubType
- CountyOrParish and MLSAreaMajor
- ListOfficeName and BuyerOfficeName (for competitive intelligence)

Skills Learned

- Feature engineering from raw fields
- Housing market analytics concepts
- Time-series variable construction

> **Deliverable – Week 6**

> Submit a .py script demonstrating all engineered metrics (price ratio, close-to-original-list ratio, PPSF, days on market, YrMo, listing-to-contract days, contract-to-close days), with a sample output table showing the new columns populated correctly. Include at least one segmented summary table grouped by PropertyType or CountyOrParish.

Week 7 – Outlier Detection and Data Quality

Extreme values in price, square footage, or days on market can distort market averages and trends. You will implement a statistical method to identify and remove these records.

Method: Interquartile Range (IQR)

The IQR method removes records that fall outside a defined statistical range:

```
Q1 = df['ClosePrice'].quantile(0.25)
Q3 = df['ClosePrice'].quantile(0.75)
IQR = Q3 - Q1
lower = Q1 - 1.5 * IQR
upper = Q3 + 1.5 * IQR
df = df[(df['ClosePrice'] >= lower) & (df['ClosePrice'] <= upper)]
```

Why This Matters

Outliers – such as a $50 million estate or a $10,000 distressed sale – can skew median prices and misrepresent the typical market. However, do not blindly remove outliers. Instead, use a tiered approach: flag extreme values first, apply business rules (e.g., ClosePrice <= 0 is always invalid), and create a separate filtered analysis dataset rather than permanently deleting records. Use percentiles alongside IQR to guide decisions.

Skills Learned

- Statistical data cleaning
- Outlier detection using IQR
- Improving dataset reliability for analytics
- Flagging vs. removing: building analysis-ready filtered datasets while preserving raw records

> **Deliverable – Week 7**
> Submit a .py script applying IQR filtering to key numeric fields (ClosePrice, LivingArea, DaysOnMarket). Add outlier flag columns rather than deleting records outright. Save both a full flagged dataset and a clean filtered dataset. Include a written comparison of dataset size and median values before and after filtering.

Weeks 8–10 – Tableau Dashboard Development

With clean, engineered datasets, you will import the data into Tableau and build interactive dashboards that tell the story of the housing market.

Deliverable – Weeks 8–10

Submit two Tableau workbooks using the cleaned, Residential-filtered dataset:

(1) market_analysis.twbx

Required dashboards (monthly, January 2024 through latest month available, all filterable by city, county, or zip code and PropertySubType):

- Monthly median close price
- Average days on market
- Average close-to-original-list price ratio
- New listings
- Closed sales

Plus one market analysis dashboard of your own design.

(2) competitive_analysis.twbx

Required dashboards (January 2024 through latest month available):

- Top 100 listing agents by sales volume and units
- Top 100 listing offices by sales volume and units — filterable by city, county, zip code, and PropertySubType
- Zip code heat map of median close prices — filterable by month, city, county, zip code, and PropertySubType
- Zip code heat map of homes sold — filterable by month, city, county, zip code, and PropertySubType

Plus one competitive analysis dashboard of your own design.

Weeks 11–12 – Final Presentation and Market Intelligence Report

The final week is dedicated to communicating your findings. You will produce a complete package: polished Tableau dashboards, a concise market intelligence report, and a live presentation.

Deliverables

Deliverable	Description
Tableau Dashboards	Publish all dashboards to Tableau Public account
1-Page Market Intelligence Report	Written summary of key market findings
5 Minute Presentation	Live walkthrough of dashboards and insights

1-Page Market Intelligence Report on any County, Zip Code, or City of Your Choice

- Market Overview – median home price trends, days on market
- Pricing Trends – price per square foot, sold-to-list ratio
- Market Activity – new listings vs. homes sold, transaction volume
- Competitive Landscape – top agents and brokerages by volume and units
- Key Takeaways – 3–5 data-driven insights discovered during analysis

Deliverable – Weeks 11–12

Submit the final two Tableau workbooks, publish the dashboards from those two workbooks to your Tableau Public account, a 1-page market intelligence report following the structure outlined above, and deliver a 5-minute live presentation walking through key findings.

Intern Expectations

To succeed in this program, interns are expected to:

- Complete each week's tasks and deliverables on schedule
- Ask questions and seek clarification when needed
- Maintain clean, well-commented Python scripts
- Document findings and observations as you progress
- Participate actively in any check-in meetings or reviews
- Treat the MLS datasets as confidential business data

Reminder: All datasets provided during this internship are sourced from live MLS transaction records. They are confidential and for program use only. Do not share or distribute this data externally.